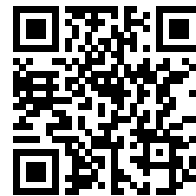


Iremide (Blessing) Oyelaja

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Education

2028 **M.S. Information Technology** – Carnegie Mellon University
2024 **B.Eng. Computer Engineering** – Federal University of Technology Akure
CGPA: 3.69/4 **First Class Honours (Top 5)**, | Awards: MTN Science and Technology Scholarship
Thesis: Comparative Performance of Single and Combined Neural Networks for Nonlinear Equalization in Optical Fiber Communication.
Selected Coursework: Linear Algebra, Calculus, Statistics, Numerical Analysis, Discrete Structures and Computer Algorithms, Artificial Intelligence, Computer Security Techniques, Operating Systems, Digital Signal Processing, Robotics and Automation etc.

Research Interests

AI Security, Privacy, Adversarial ML, Vision Language Models, Multimodal AI

Publications

- [1] Obiefuna N.*, Oyeneye S.*, Odunaiya S.*, **Oyelaja I.***, Kolawole S.* "Privacy Isn't Free: Benchmarking the Systems Cost of Privacy-Preserving ML". *Equal Contribution. *ICML 2025 Workshop on Efficient Systems for Foundation Models (ES-FoMo)*. — Code
- [2] Obiefuna N., **Oyelaja I.**, Odunaiya S., Oyeneye S. "Secure and Scalable Horizontal Federated Learning for Bank Fraud Detection". *ICLR 2025 Workshop on Financial AI*. — Code
- [3] Dahunsi F., **Oyelaja I.**, Ponnle A., Adegoke M. "The Bane of Nigerian Industries and Households: The Quality of Energy". *FUOYE Journal of Engineering and Technology*, Volume 10, Issue 2, June 2025.

Research Experience

- 2024 – **ML Research Engineer**, ML Collective
- Present
- Revealed **limitations in PPML performance** by demonstrating that privacy techniques (FL, DP, SMPC) interact in non-additive and sometimes catastrophic ways, challenging the widely assumed modularity in deployment.
 - Discovered **fundamental incompatibilities in FL+DP**: traced failures to signal-to-noise collapse, resulting in accuracy dropping from **98% to 13%** and energy costs rising over 20×.
 - Demonstrated FL+SMPC stability and efficiency: preserved model accuracy with minimal overhead, establishing the **importance of abstraction layer** alignment in privacy system design.
 - Validated HFL as a regulation-compliant, scalable alternative to centralized training for fraud detection; presented insights on secure collaboration in sensitive financial environments.
 - Benchmarked model overheads: **identified 98.9% higher bandwidth usage** for transformers versus MLPs (138.6MB vs. 1.4MB) and a **4× increase** in training time, highlighting practical deployment challenges.
- 2021 – 2024 **Undergraduate Research Intern**, Smart Systems Research Laboratory, Akure
- Conducted cross-utility infrastructure analysis to **identify vulnerabilities** and **regional discrepancies**, informing risk mitigation strategies and enhancing smart meter security.
 - Performed load forecasting based on **extrapolation and correlation** to optimize energy distribution in microgrids.
 - Applied load volatility analysis (PAR > 80) to identify sharp consumption spikes, informing infrastructure stress points and supporting demand-side intervention planning.

Technical Skills

Languages: Python, C/C++, MATLAB, Bash | **Frameworks:** Pytorch, Tensorflow, Transformers, Flower, Opacus, Node.js, Flask, Differential Privacy (DP-SGD, LDP) | **Tools:** Git, VSCode, CI/CD, W&B, GCP, LaTeX, MCP

Conference Presentations

2025	Poster Presenter , Deep Learning Indaba, Kigali, Rwanda <i>Secure and Scalable Horizontal Federated Learning for Bank Fraud Detection</i>
2024	Poster Presenter , Deep Learning Indaba, Dakar, Senegal <i>Comparative Performance of Single and Combined Neural Networks for Nonlinear Equalization in Optical Fiber Communication</i>

Selected Projects

2026	MotoGuide , Built an AI-assisted navigation system designed to overcome language barriers and support destination routing and fare estimation for new immigrants in Kigali. Winner of CMU Tech SIP 2026.
2025	LearNEXO , Designed a recommendation system that personalizes learning content using click-stream behavior, quiz performance, and session history to adapt difficulty and sequence for each learner.
2025	Spot’d , Built a lightweight user verification pipeline that combines social login, device fingerprinting, and real-time engagement signals to flag inauthentic accounts before they spread content.
2025	LoanGuard , Built a privacy tool that detects predatory terms and conditions before users sign up.
2023	Fake/Real Job Tracker , Developed an NLP pipeline that classifies job postings as real or fraudulent using crowdsourced labels and domain-tuned classifiers, reducing false positives on held-out samples through iterative error analysis.

Awards

2026	Winner, Tech SIP , Carnegie Mellon University (MotoGuide)
2025	Best Poster Award , Deep Learning Indaba, Kigali, Rwanda
2024	Award of Academic Excellence , Federal University of Technology Akure
2024	Best Poster Award , Deep Learning Indaba, Dakar, Senegal
2024	Winner of Connected Black Professional Ideathon , Cisco
2023	KaggleX BIPOC Fellowship , Kaggle
2020	MTN Science and Technology Scholarship , MTN

Leadership and Volunteering Experience

2023 – 2024	Community Lead , Data Science Nigeria. <i>A 500+ member non-profit organization focused on building AI literacy across Africa.</i> • Recognized and awarded as the most active Data Science Nigeria community in 2024. • Designed and delivered Python and data science tutoring programs for 500+ undergraduates, using hands-on projects to grow a local talent pipeline.
2022 – 2024	Software Engineering Team Lead , Smart Systems Research Laboratory, Akure. <i>University Research Group focused on AI, IoT and Embedded Systems.</i> • Managed cross-functional teams of 6 undergraduate interns, overseeing research projects from conception to successful implementation.
2023	Cybersecurity Team Lead , Department of Computer Engineering. <i>Student-led Software Security Auditors.</i> • Conducted security audits of a complex web application with high-impact vulnerabilities and proposed architectural fixes that were recognised by faculty as one of the strongest projects in the cohort.
2020 – 2021	Data Science Instructor , Google Developers Group, Akure. <i>A Local Community of over 100 Developers.</i> • Tutored 15 community members in Python and data science fundamentals concepts in biweekly meetups through structured lessons and personalized guidance. • Demystified core concepts and created an inclusive, engaging learning environment that encouraged hands-on practice.